

Outcomes of congenital diaphragmatic hernia repair on extracorporeal life support

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Abstract

Purpose Extracorporeal life support (ECLS) is applied to refractory pulmonary hypertension in congenital diaphragmatic hernia (CDH). We evaluate the single-center outcomes of infants with CDH to determine the utility of late repair on ECLS versus repair post-decannulation.

Methods Records of infants with CDH (2004–2014) were retrospectively reviewed.

Results CDH was diagnosed in 177 infants. Sixty six (37%) underwent ECLS, of which, 11 died prior to repair, 33 were repaired post-decannulation, and 22 were repaired on ECLS. Repair was delayed in patients on ECLS (19 versus 10 days, $p < 0.001$). Patients repaired on ECLS had longer ECLS runs (22 versus 12 days, $p < 0.001$) and higher rates of bleeding and mortality than those repaired post-decannulation. Survival was 54% in infants undergoing ECLS, 65% in those who underwent repair, 36% in those repaired during ECLS, and 85% in those who were decannulated prior to repair. Eighteen percent ($N = 4$) of deaths after repair on ECLS were attributable to surgical bleeding. The remainder was due to pulmonary hypertension or sepsis.

Conclusion Infants who underwent CDH repair post-decannulation had excellent outcomes and no mortalities attributable to repair. Neonates who underwent repair on ECLS late on bypass had the lowest survival rate with only 18% of mortality in this cohort attributable to surgical bleeding.

Keywords Congenital diaphragmatic hernia · Extracorporeal life support · Post-operative bleeding · Pulmonary hypertension

Introduction

Congenital diaphragmatic hernia (CDH) affects 1 in 3000 live births and has considerable associated morbidity and mortality [1, 2]. There is no consensus on clinical management, in part due to the wide variability in disease severity [3, 4]. Current thought is that pulmonary hypoplasia (PH) and persistent pulmonary hypertension (PPHN) are the most significant factors determining survival [5, 6]. Therefore, treatment of CDH is primarily directed towards management of PH and PPHN followed by surgical repair to restore normal anatomy after cardiorespiratory stabilization [7, 8].

Extracorporeal life support (ECLS) has been utilized since the 1970's to stabilize patients with PH and PPHN that do not respond to medical management [9, 10]. The use of ECLS for long-term oxygenation and perfusion in neonates with CDH has allowed high-risk patients to stabilize and ultimately survive [2, 10–12]. However, CDH patients who require ECLS continue to have lower survival and higher complication rates than their non-ECLS counterparts [13, 14]. This is due to their inherent disease severity as well as the additional risks of ECLS, including bleeding, embolization, renal dysfunction, and poor neurologic outcomes [14–18]. Moreover, as medical management has improved over time, clinicians are progressively selecting a more critically ill cohort for ECLS support. In turn, this may contribute to longer ECLS episodes, more accrued complications and mortality, and increasingly frequent difficulties liberating patients from ECLS.

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Although timing of diaphragmatic repair in these patients remains controversial, clinicians generally believe that CDH repair on ECLS has significant associated risks primarily related to anticoagulation [4]. Risks of repair on ECLS may include hemorrhage, hemodynamic instability, and abdominal compartment syndrome [4, 7, 16, 19, 20]. Some institutions advocate for the early repair prior to ECLS cannulation, though there is limited evidence to support emergent CDH repair prior to stabilization in all patients [21–23]. Indeed, the cohort of children who undergo repair of CDH after ECLS decannulation has excellent outcomes after stabilization and without anticoagulation [5, 24]. However, there remain a notable percentage of infants placed on ECLS for CDH who will expire while on bypass without operative intervention or who are deemed “unweanable” from ECLS. It is this sub-group that raises concerns that intervention prior to ECLS, at initiation, or as a salvage maneuver may lead to improved survival in this sub-group. Unfortunately, there is no accurate model to identify which patients will stabilize on ECLS allowing for post-decannulation repair and which infants will benefit from the late repair on ECLS [4, 21, 22] and there is no clear cutoff for how long patients should remain on ECLS prior to being deemed “unweanable”.

Current practice at our institution is to support patients with refractory PH and PPHN on ECLS, decannulate, and perform operative repairs at a variable time interval off bypass. However, for infants who cannot be liberated from ECLS, definitive repair may be considered according to the individual attending Pediatric Surgeon and Neonatologist, while the infant remains on ECLS. Lack of uniformity of approach stems from variable standards for ECLS termination, diverse experience with operation on ECLS, and lack of data regarding the safety and efficacy of late repair of CDH on ECLS. Here, we evaluate all CDH repairs performed at a single free-standing pediatric hospital and focus on outcomes in the subset of infants who are deemed unable to wean from ECLS and undergo late repair of their CDH while on bypass.

Methods

Following IRB approval, records of infants (0–30 days) presenting to a single pediatric hospital from 2004 to 2014 with a diagnosis of CDH were retrospectively reviewed. Demographics, co-morbidities, operative records, ECLS use, complications, and outcomes were evaluated. The center is the largest, free-standing children’s hospital in its geographic region and has been operating a neonatal ECLS program continuously since 1988.

Statistical analysis was performed with the GraphPad Prism 6.0 student *t* test with Bonferroni correction for

multiple comparisons, Mann–Whitney, and Kruskal–Wallis tests where appropriate. Data are represented as mean $\pm$ standard deviation (SD) unless otherwise specified.

Results

Patient population and demographics

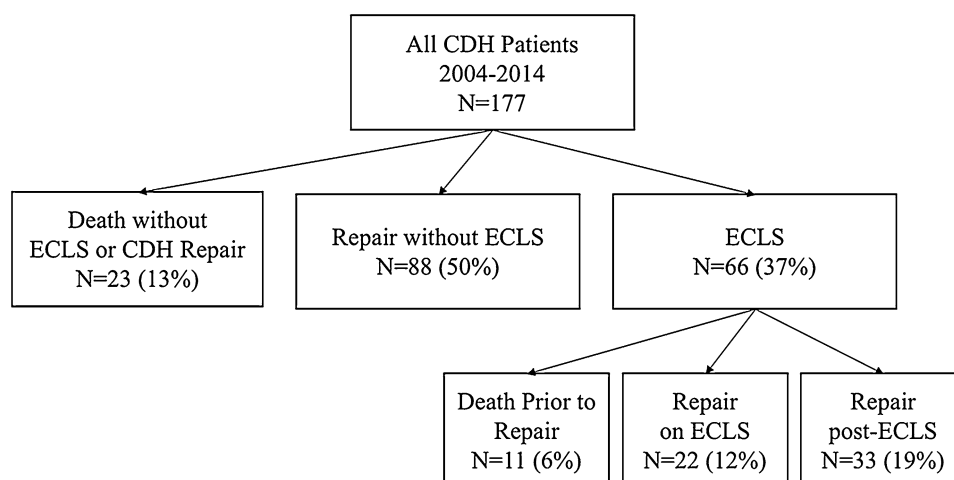
CDH was diagnosed in 177 infants during our study period. Repair was performed without the use of ECLS in 88 patients. Twenty-three patients died prior to CDH repair without ECLS due to associated anomalies ($N = 10$), death on arrival ($N = 3$), prematurity ($N = 2$), or were made comfort care ($N = 8$) and were excluded from further analysis. ECLS was used in 66 (37%) patients; of which, 11 died prior to repair, 33 were repaired post-decannulation, and 22 were repaired on ECLS (Fig. 1). Of patients who survived to CDH repair or ECLS cannulation, 62% were male, 45% had a prenatal diagnosis of CDH, 27% were premature (<38 weeks gestational age), and the average birthweight was 3105 ± 579 g.

Infants who required ECLS compared with infants did not require ECLS

We evaluated differences in patient demographics, management, and outcomes for patients who underwent ECLS versus those that did not require ECLS (Table 1). There was no significant difference in gender, prematurity, or birthweight between groups. Patients who required ECLS were more likely to have a prenatal diagnosis (62.9 versus 33%, $p < 0.001$) and have liver in their diaphragmatic defect (66 versus 20%, $p < 0.001$). In addition, patients who required ECLS were significantly older at CDH repair (18.7 ± 7.2 versus 10 ± 7.3 days old) and had significantly higher rates of bleeding complications (25.5 versus 0%) and mortality (44 versus 1.1%).

Outcomes in patients who underwent ECLS

Patients who underwent ECLS for stabilization of PH and PPHN were evaluated. Average ECLS duration was 16 ± 8.9 days (range 1–52 days). Of 66 patients who were placed on ECLS, 11 patients died prior to surgical repair. Of those patients, eight patients were made comfort care due to severe PPHN and failure to improve on ECLS, one was made comfort care due to a lethal genetic anomaly, and two developed intraventricular hemorrhage on ECLS. The average duration of ECLS in this group was 16 ± 11 days (range 1–38 days).

Fig. 1 CDH patient population

Table 1 Demographics, management, and outcomes of patients who required ECLS compared with those that did not require ECLS

	No ECLS (N = 88)	ECLS (N = 66)	p value
Gender (% male)	64.8%	59%	0.50
Pre-term birth (<38 weeks) (%)	31.8%	21.5%	0.20
Birthweight (g ± SD)	3082 ± 640	3136 ± 488	0.57
Prenatal diagnosis (%)	33.0%	62.9%	0.001
Left CDH (%)	87.5%	76.2%	0.08
Liver in defect (%)	20%	66%	0.001
DOL at CDH repair (days ± SD)	10 ± 7.3	18.7 ± 7.2	0.001
Bleeding complications (%)	0%	25.5%	0.001
Mortality (%)	1.1%	44%	0.001

ECLS extracorporeal life support, DOL Day of life, CDH congenital diaphragmatic hernia

Fifty-five patients who were placed on ECLS underwent CDH repair. Of those, 22 (40%) were unable to be weaned off of ECLS prior to repair. Repair on ECLS was performed on day of life 18.9 ± 6 (range 5–30 days). When compared with infants who were repaired following ECLS decannulation ($N = 33$, 60%), infants who were repaired on ECLS had similar rates of prematurity, timing of repair, and liver in their diaphragmatic defect. Infants repaired on ECLS had a significantly longer ECLS duration (22.2 ± 10 versus 11.8 ± 4 days), higher rate of post-operative bleeding (50 versus 6%), and higher mortality (64 versus 15%). There was no difference in veno-venous (VV) versus veno-arterial (VA) ECLS cannulation rates between groups; VA ECLS was used in 89% of patients (Table 2).

Overall, survival was 54% in infants who underwent ECLS, 65% in those who survived to CDH repair, 36% in those operated upon late in the ECLS run, and 85% in those who were decannulated prior to repair. Of the 5 out of 33 infants who died following CDH repair post-ECLS decannulation, 3 died secondary to severe PPHN on day of life (DOL) ranging from 21 to 36, 1 died secondary to PPHN and sepsis on DOL 67, and 1 died at 4 months of

age of unrelated causes. No patients who had a CDH repair after decannulation died of complications directly attributable to the operations. While 14 out of 22 patients died following CDH repair on ECLS, only 4 (18%) of neonatal deaths in those who underwent repair on ECLS were attributable surgical site bleeding. The remainder of the mortality in this group was accounted for by severe pulmonary hypertension or sepsis late in their hospitalization.

Discussion

While ECLS is frequently used for the management of refractory pulmonary hypertension in CDH, the duration of ECLS therapy and the optimal timing of CDH repair remain controversial. Preoperative stabilization followed by delayed surgical repair results in excellent outcomes and has become a routine and standard practice [8, 25]. Some centers have recently advocated for the early repair of CDH in patients on ECLS or prior to ECLS citing normalization of anatomy, decreased mass effect, improved venous

Table 2 Characteristics of infants who required ECLS support prior to CDH repair

	Death prior to repair (<i>N</i> = 11)	Repair post-decannulation (<i>N</i> = 33)	Repair on ECLS (<i>N</i> = 22)	<i>p</i> value
Pre-term birth (<38 weeks)	5 (45%)	5 (15%)	5 (23%)	0.13
DOL at CDH repair (days $\pm$ SD)	N/A	18.6 $\pm$ 8	18.9 $\pm$ 6	0.89
ECLS duration (days $\pm$ SD)	16.0 $\pm$ 11	11.8 $\pm$ 4	22.2 $\pm$ 10	0.001
VA:VV ECLS	10:1	29:4	20:2	0.9
Liver in defect	N/A	20 (61%)	14 (64%)	0.99
Post-operative bleeding complication	N/A	2 (6%)	11 (50%)	0.001
Mortality	11 (100%)	5 (15%)	14 (64%)	0.001
Post-operative day of mortality $\pm$ SD	N/A	41 $\pm$ 45	19 $\pm$ 35	0.27
Mortality secondary to CDH repair	N/A	0/5 (0%)	4/14 (18%)	0.29

DOL day of life, *VA* veno-arterial, *VV* veno-venous

return, and decreased anasarca at time of operation [21–23], yet current supporting data are limited. The practice of the early repair of all infants on ECLS may subject a large subset of patients to an unnecessary increased risk of post-operative complications that would otherwise be avoided by delaying operation until after decannulation. Urgent repair prior to cannulation and prior to hemodynamic and pulmonary stability could increase the need for application of ECLS in patients who may not have required it with the conventional timing of operation [23]. While the utility of the early repair on or prior to ECLS remains unclear, CDH repair post-decannulation has excellent outcomes [5, 24]. Some patients are unable to be weaned off of ECLS prior to CDH repair, raising the question whether operation while on bypass may allow them to be liberated from extracorporeal support. However, it is worthy of note that clinicians do not have consensus on criteria for liberating CDH infants from ECLS. Therefore, it is possible that a sub-group of these children may be considered for operative intervention on circuit that might otherwise have been weaned and operated upon in a delayed fashion. This monograph does not explicitly evaluate this possibility and assumes, for the purposes of analysis, that all infants who were operated upon late in the ECLS episode were, in fact, incapable of coming off of ECLS prior to surgical intervention. We demonstrate that infants repaired post-decannulation have excellent outcomes and that those infants repaired on ECLS have a much lower yet non-negligible survival rate and rarely die secondary to complications of late CDH repair.

To compare our patients to the known CDH population, we evaluated all infants who presented to our institution with a diagnosis of CDH. Our hospital is a free-standing children's hospital with all infants as outborn transfers from another hospital. This outborn patient population

often leads to a pre-referral selection bias resulting in a high proportion of severe CDH patients requiring ECLS (37%) when compared with the reported CDH population (7–33%) [24–28]. Despite a complex patient population, 69.5% of all CDH patients at our institution survived which is similar to data in the CDH Study Group registry [29]. CDH repair was performed in 80.8% of our patients with an overall survival rate of 74%.

To determine what factors may have affected our outcomes and mortality, we evaluated patients who died without ECLS or CDH repair. While these patients were excluded from our analysis, they comprised 13% of our overall population. The majority of these patients had associated anomalies or genetic disorders that affected patient outcome followed by patients who were made comfort care following multidisciplinary discussions with the infant's families. A small group of patients were not ECLS candidates due to prematurity or died on arrival to our hospital. A recent study utilizing CAPSNet data had similar findings in patients who died without CDH repair [26].

In our population, 37% of infants were placed on ECLS with an overall survival of 54%. Infants who were placed on ECLS and survived to CDH repair had an overall survival of 65%. This is comparable to outcomes in the CDH Study Group which reported overall survival of 48% in all children with CDH receiving ECLS support and 61% in those who underwent surgical repair [29]. There was no difference in VA versus VV ECLS between our study groups and VA was used in 89% of patients. The previous studies have found no difference in outcomes between VA and VV ECLS cannulation in CDH patients [20, 30]. Timing of repair was similar in those repaired on ECLS (DOL 18.6 $\pm$ 8) and those repaired post-decannulation (DOL 18.6 $\pm$ 8) but was significantly later than infants

who did not require ECLS (DOL 10 ± 7.3). Average ECLS duration was 16 ± 8.9 days (range 1–52 days) and was significantly higher in infants who died prior to repair and in those who were repaired on ECLS. Interestingly, Tiruvoipati et al. found only an 18% survival in infants who required ECLS support for longer than 14 days which was significantly lower than those who required less than 14 days (68%) [31]. Despite an increased mortality reported with longer ECLS runs, improvement in pulmonary function allowing decannulation can take 4 weeks or longer [14]. Current recommendations from the American Pediatric Surgical Association outcomes and evidence-based practice committee state that durations up to 4 weeks may still be associated with acceptable survival [20]. Duration of ECLS support in patients with severe PPHN and PH requires further evaluation.

It is critical to determine if the difference in mortality between patient groups was secondary to disease severity or timing of surgical repair. Not surprisingly, there was significantly lower mortality in patients who did not require ECLS (1.1%) compared with those who were placed on ECLS (44%) likely due to differences in underlying disease severity between the two groups. However, the cause of mortality in patients who underwent a CDH repair on ECLS or post-decannulation may be more complicated. Here, we report a 15% mortality in infants repaired following decannulation, yet no patient in this group died of complications directly attributable to the operations. We report a 64% mortality in infants who required CDH repair; while on ECLS, however, only 4 of 14 deaths were related to surgical site bleeding. This suggests that while there is a significantly higher risk of post-operative bleeding complications in infants repaired while on ECLS compared with post-decannulation, the high mortality is attributable to other causes not specifically identified here. An increased risk of bleeding complications while patients are anticoagulated on ECLS has been reported previously and is an inherent risk of repair on ECLS [5, 7, 16]. More likely causes of elevated mortality risk in the group of children repaired at the end of ECLS therapy is selection of patients with more severe pulmonary hypoplasia, pulmonary hypertension, cardiovascular dysfunction, or complications of more prolonged extracorporeal support.

While we did not have any patients who were repaired within the first 72 h of cannulation, we compared our data to published reports of the “early” repair. Dassinger et al. reported 34 patients who underwent repair on ECLS at an average of 55 h after cannulation with 71% survival and 8.8% surgical site hemorrhage [22]. Fallon and colleagues reported that 11 patients who were repaired on ECLS within 72 h of cannulation with a 73% survival rate with 36% of infants have a major surgical bleeding complication. These results may compare favorably to the aggregate

survival rate of 54% seen in our series. It is possible that the increased risk to infants who would have otherwise survived to ECLS decannulation and delayed repair [5, 24] is balanced by increased survival in those who would have expired without operation or would have been operated upon later in the ECLS episode. Additional information on late mortality and disability will be needed to discern whether any possible aggregate population benefit is durable and does not contribute to diminished functional outcomes. Moreover, additional studies are needed to determine predictors of failure to wean from ECLS, better outcome with the early or late operation, or which patients can be reasonably predicted to be liberated from ECLS. Currently, there are no known variables to identify patients who will stabilize on ECLS prior to CDH repair [4, 21, 22]. In our study, infants who underwent repair of CDH after decannulation had excellent outcomes and no mortalities secondary to surgical repair. Neonates who were unable to wean off ECLS and underwent repair on ECLS had a low but non-negligible survival rate where the majority of deaths were unrelated to post-operative complications. These findings suggest that there is a widely diverse set of physiologies and performances of infants with CDH that will benefit from improved risk stratification and application of different management strategies from the early operation before ECLS, the early operation on ECLS, the late operation on ECLS, and delayed repair with or without ECLS.

There are limitations to our study as it is a retrospective single-center review over a period of 11 years. During this time frame, the medical management of patients with CDH has evolved. This is particularly notable when the primary endpoint of CDH studies is mortality. Indeed, as is seen here, much of the mortality in infants with CDH appears to result from complications of pulmonary hypoplasia and pulmonary hypertension. However, improved and standardized approaches to delivery room care, ventilatory management, sedation, and analgesia, and fluid administration may have impacts that cannot be quantified in a retrospective study. In addition, selection bias may contribute significantly to the results reported here. Each pairing of Neonatologist and Pediatric Surgeon in our institution has discretion to determine the timing of operative intervention as well as the initiation, management, and liberation from ECLS. Therefore, which group an infant was in may be a function not just of physiology and performance, but the individual biases of the care team members with respect to the safety and effectiveness of operative intervention before, during, or after ECLS. In the absence of data or consensus guidelines, we can only draw associations based upon these biases and more rigorous study would be needed to best define patient selection for surgical timing.

Despite these limitations, we demonstrate that infants repaired following ECLS decannulation have excellent outcomes and those who undergo repair on ECLS towards the end of the run have an acceptable survival rate and rarely die from post-operative bleeding complications. This suggests that a possible utility of delayed surgical repair during ECLS for CDH in whom the care team believes liberation from ECLS is otherwise not possible.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflicts of interest.

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